Vinay Kumar Thota

Healthcare Data Analyst

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Healthcare Data Analyst with 3+ years of experience in provider and life sciences settings, specializing in EHR integration, claims analytics, and compliant data modeling. Built HL7/FHIR-enabled workflows and optimized warehouses supporting multi-million record clinical datasets with 99.6% accuracy. Developed risk-adjusted models and automated quality metrics, improving reporting efficiency by 26% and accelerating clinical decisions by 31%. Experienced in population health, RWE studies, and CMS-aligned reporting. **Certified in Data Analysis with R Programming, Data Analysis with Python, and Python Data Analytics, with expertise in EHR analytics, Power BI, Tableau, and Excel.**

PROFESSIONAL EXPERIENCE

Healthcare Data Analyst | Steer Health **Oct 2024 – Present**

- Architected scalable ETL pipelines using Python and SSIS to ingest multi-facility EHR feeds from MEDITECH, improving structured data availability for downstream analytics by 32% across clinical data marts.
- Engineered interoperability workflows leveraging HL7 messaging standards and FHIR APIs integrated with Cerner, enhancing real-time exchange of patient encounters and reducing interface latency by 27%.
- Modeled longitudinal patient datasets in PostgreSQL and optimized warehouse performance in AWS Redshift, enabling advanced survival analysis using R (tidyverse, ggplot2) and reducing complex query runtime by 38%.
- Developed executive-level dashboards in Power BI utilizing advanced DAX measures to track clinical KPIs including readmission rates, LOS, utilization, and patient outcomes, driving a 21% improvement in care performance monitoring efficiency.
- Consolidated medication normalization using RxNorm and laboratory standardization via LOINC within the Sentinel Common Data Model, strengthening data consistency for population health and risk adjustment modeling initiatives.
- Implemented secure data lake architecture using AWS S3 and ensured regulatory data integrity compliance aligned with FDA 21 CFR Part 11, supporting audit traceability and zero critical findings during compliance assessments.
- Performed payer benchmarking and utilization analytics using Optum Analytics datasets while extracting ambulatory care data from eClinicalWorks, leading to a 17% reduction in avoidable readmission rates through risk-adjusted outcome modeling.

Associate Healthcare Data Analyst | Cipla **Sep 2021 – Jul 2023**

- Extracted and validated EHR datasets from Epic and research data in REDCap to support RWE studies and cohort analysis while ensuring accurate data reconciliation and compliance with ICH-GCP, improving validation accuracy by 24%.
- Led and optimized claims analysis queries in MySQL within athenahealth to evaluate CPT and ICD - 10 coding accuracy and reimbursement trends while enhancing RCM reporting workflows, contributing to a 19% reduction in claim denial rates.
- Performed data cleansing and exploratory analysis using Python and built KPI dashboards in Tableau to monitor enrollment, protocol deviations, and turnaround time, reducing reporting cycle time by 28% through structured automation.
- Standardized clinical datasets per CDISC SDTM guidelines and managed EDC validation workflows in Medidata Rave while conducting statistical quality checks using SAS to ensure audit-ready regulatory submissions.
- Coordinated multi-source clinical and payer datasets using Health Catalyst and benchmarked analytics against Optum Insight datasets while enforcing PHI protection under HIPAA governance frameworks.
- Supported study lifecycle data capture in OpenClinica and assisted ETL workflows in Google Cloud Platform (GCP) while contributing to compliance analytics unified with Joint Commission standards, achieving 100% audit-ready internal reviews.

PROJECTS

Population Health Analytics & Chronic Disease Progression Modeling

- Built a longitudinal risk stratification model on multi-year EHR and claims data for chronic diseases, achieving 84% AUC and reducing simulated avoidable ER visits through high-risk cohort identification.
- Configured patient segmentation and care-gap analytics using demographic, adherence, and lab trend variables, improving preventive screening compliance and care prioritization efficiency by 28%.

Tech Stack: Python, SQL, SAS, Tableau, REDCap, HL7/FHIR

Clinical Quality Metrics Automation & Regulatory Compliance Analytics

- Automated HEDIS and CMS-mapped quality metric calculations from encounter and diagnosis datasets, reducing manual reporting and improving accuracy by 19%.
- Designed compliance and coding anomaly detection models to flag documentation gaps, increasing simulated provider benchmark performance by 17%.

Tech Stack: SQL Server, Python, Power BI, Cerner Millennium (synthetic data), ICD-10/CPT, AWS S3

EDUCATION

M.S. in Health Data Analytics, University of North Texas, USA **Aug 2023 – May 2025**

Bachelor of Pharmacy, Acharya Nagarjuna University, India **Sep 2019 – Apr 2023**

TECHNICAL SKILLS

- **Clinical & Healthcare Systems:** MEDITECH, Cerner, Epic, eClinicalWorks, athenahealth, Optum Analytics, Optum Insight, Health Catalyst
- **Healthcare Data Analytics & Statistical Analysis:** Python, SQL, SAS, R (tidyverse, ggplot2), Survival Analysis, Risk Stratification Modeling, Cohort Analysis, Real-World Evidence (RWE) Analytics, Population Health Analytics, Quality Metrics Automation
- **Business Intelligence & Clinical Reporting:** Power BI (DAX), Tableau, HEDIS, CMS Quality Measures
- **Healthcare Interoperability & Clinical Data Standards:** HL7, FHIR APIs, RxNorm, LOINC, Sentinel Common Data Model, CDISC SDTM
- **Databases & Healthcare Data Warehousing:** PostgreSQL, MySQL, SQL Server, AWS Redshift
- **ETL & Healthcare Data Processing:** SSIS, ETL Pipeline Development, Data Validation, Data Cleansing, Data Reconciliation, AWS S3, Google Cloud Platform (GCP), Secure Data Lake Architecture
- **Healthcare Coding & Regulatory Compliance:** ICD-10, CPT, ICH-GCP, HIPAA, FDA 21 CFR Part 11, Joint Commission Standards

CERTIFICATION

- Data Analysis with R Programming
- Data and Electronic Health Records
- Harnessing the Power of Data with Power BI
- Getting Started with Tableau
- Meta Data Analyst
- Preparing Data for Analysis with Microsoft Excel
- Data Analysis with Python
- Python Data Analytics
- Data Analysis with Spreadsheets and SQL